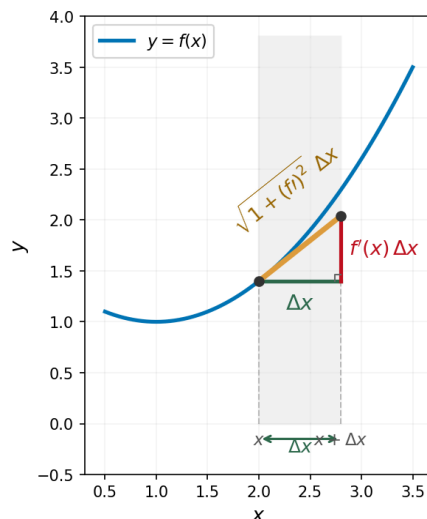


From Arc Length to Surface Area

In Calc I, you learned that the **arc length** of $y = f(x)$ from a to b is:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Where does this $\sqrt{1 + (f')^2}$ come from?



Zoom in on one small piece. The curve segment has length $\approx$ the **hypotenuse**, not the horizontal distance Δx . By the Pythagorean theorem:

$$\text{length} \approx \sqrt{(\Delta x)^2 + (f' \Delta x)^2} = \sqrt{1 + (f')^2} \Delta x$$

The factor $\sqrt{1 + (f')^2}$ is a **correction factor** that accounts for the **tilt** of the curve. If $f' = 0$ (flat), the correction is 1. If the curve is steep, the correction is larger.

Surface area is the same idea — but in two dimensions. We need a correction factor that accounts for the tilt of a **surface**, not just a curve.

Arc length: $\int \sqrt{1 + (f')^2} dx$. Surface area: $\iint \sqrt{?} dA$. Our job is to find what goes under the square root.

Why Surface Area Needs a Correction Factor

Think of draping a **bedsheet over a tent pole**. The sheet covers more area than its shadow on the floor.

A tilted or curved surface has **more area** than its projection onto the xy -plane. How much more depends on how **steep** the surface is.

For a flat region D in the xy -plane, area = $\iint_D 1 dA$.

For a curved surface $z = f(x, y)$ above D :

$$\text{surface area} = \iint_D (\text{correction factor}) dA$$

The correction factor is ≥ 1 everywhere. It equals 1 only where the surface is perfectly flat ($f_x = 0$ and $f_y = 0$).

We need to find the correction factor for surfaces. The arc length correction was $\sqrt{1 + (f')^2}$. The surface area correction will involve **both** partial derivatives f_x and f_y .

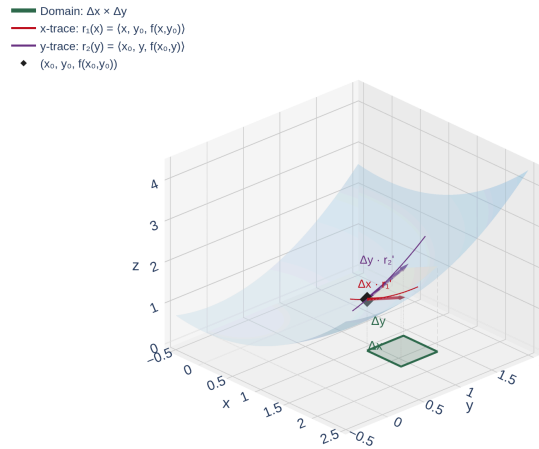
Deriving the Correction Factor

Let's carefully derive the area of one small surface patch. Consider $z = f(x, y)$ above a small rectangle $[x_0, x_0 + \Delta x] \times [y_0, y_0 + \Delta y]$.

Step 1 — Parameterize two traces through the corner point $(x_0, y_0, f(x_0, y_0))$:

- **x-trace** (hold $y = y_0$ fixed): $\vec{r}_1(x) = \langle x, y_0, f(x, y_0) \rangle$
- **y-trace** (hold $x = x_0$ fixed): $\vec{r}_2(y) = \langle x_0, y, f(x_0, y) \rangle$

These are curves **on the surface**.



[Interactive version](#)

Step 2 — Tangent vectors to the traces:

$$\vec{r}'_1(x) = \langle 1, 0, f_x \rangle \quad \vec{r}'_2(y) = \langle 0, 1, f_y \rangle$$

Step 3 — Approximate the patch sides. You already know that displacement $\approx$ velocity $\times \Delta t$. Here Δx and Δy play the role of Δt (the change in the parameter):

$$\vec{a} = \vec{r}'_1 \Delta x = \langle \Delta x, 0, f_x \Delta x \rangle \quad \vec{b} = \vec{r}'_2 \Delta y = \langle 0, \Delta y, f_y \Delta y \rangle$$

Step 4 — Area of the parallelogram via the cross product:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \Delta x & 0 & f_x \Delta x \\ 0 & \Delta y & f_y \Delta y \end{vmatrix} = \langle -f_x, -f_y, 1 \rangle \Delta x \Delta y$$

$$|\vec{a} \times \vec{b}| = \sqrt{f_x^2 + f_y^2 + 1} \Delta x \Delta y = \sqrt{f_x^2 + f_y^2 + 1} \Delta A$$

Sum over all patches and take the limit:

$$A(S) = \iint_D \sqrt{f_x^2 + f_y^2 + 1} dA$$

The correction factor $\sqrt{f_x^2 + f_y^2 + 1}$ comes directly from the cross product of the tangent vectors to the surface traces. This same construction — parameterize, take partials, cross product — will reappear in Chapter 16 for surface integrals over parametric surfaces.

The Surface Area Formula

Summing up the parallelogram areas and taking the limit gives us the integral formula.

Surface Area

The area of the surface $z = f(x, y)$ over a region D , where f_x and f_y are continuous, is

$$A(S) = \iint_D \sqrt{[f_x(x, y)]^2 + [f_y(x, y)]^2 + 1} \, dA$$

Sanity check: If f is **constant** (a flat horizontal surface), then $f_x = 0$ and $f_y = 0$, so the integrand is $\sqrt{0 + 0 + 1} = 1$, and we get $A(S) = \iint_D 1 \, dA = \text{area of } D$. That's exactly right — a flat surface has the same area as its shadow!

The formula $\sqrt{f_x^2 + f_y^2 + 1}$ always gives a value ≥ 1 , so the surface area is always at least as large as the area of D . Equality only when the surface is perfectly flat.

Example 1: Plane Inside a Cylinder

Example 1

Find the surface area of the part of the plane $2x + 5y + z = 10$ that lies inside the cylinder $x^2 + y^2 = 9$.

Solve for $z = f(x, y)$, compute the partials, and set up the integral over the disk. What shape is D ?

Example 2: Area of a Spherical Cap

Example 2

Find the surface area of the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the plane $z = 1$.

Solve for z (upper hemisphere), find f_x and f_y , simplify the integrand, then switch to polar. What is the domain D ?

Example 3: Why $A(\text{sphere}) = 4\pi a^2$

Example 3

Prove that the surface area of a sphere of radius a is $4\pi a^2$.

Strategy: Find the area of the upper hemisphere $z = \sqrt{a^2 - x^2 - y^2}$ and double it.

Set up the integrand for $z = \sqrt{a^2 - x^2 - y^2}$ over the disk $x^2 + y^2 \leq a^2$. Use the pattern from Example 2.

Homework

Section 15.5: 1, 3, 9, 11, 13, 23